

Quantifying the Impact of NBA Hustle Plays on Regularized Adjusted Plus-Minus

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Introduction

The intangible aspects of basketball, such as hustle and effort, are often used to evaluate players on the eye-test at all levels of play. Starting in the 2015-16 season, the National Basketball Association started tracking these hustle plays. Quantifiable metrics like deflections, contested shots, screen assists, loose balls recovered, and box outs have brought workable data to these standout plays. At the same time, advanced analytics has continued to evolve measures of player impact. Regularized Adjusted Plus-Minus is one such metric that isolates a player’s overall effect on team performance, independent of teammates or opponents. This research seeks to bridge the eye-test with advanced analytics, observing and quantifying the relationship between hustle and RAPM.

Methodology

Data Collection

- Collected publicly available hustle stats from the NBA
 - Selected deflections, contested shots, screen assists, loose balls recovered, and box outs.
- 5-Year 6 Factor RAPM from databallr.
 - Breaks a player's impact into offensive and defensive efficiency, turnovers, and rebounding, showing how many points per 100 possessions they add in each area over a 5Y period.

Pre-Processing

- Converted all hustle stats to per-36 minutes to normalize playing time, filtered fringe players by setting minimum minutes at 500 per season.
- Computed 5-year averages (2019-20 through 2023–24) for each player to align with 5-Year RAPM.

Metric Construction

- Use a linear regression model to extract data-driven weights, combining standardized hustle stats into a single composite: Hustle Index.

$$HI_i = \beta_1 \cdot Z(\text{Deflections}_i) + \beta_2 \cdot Z(\text{Contested Shots}_i) + \beta_3 \cdot Z(\text{Screen Assists}_i) + \beta_4 \cdot Z(\text{Loose Balls}_i) + \beta_5 \cdot Z(\text{Box Outs}_i)$$

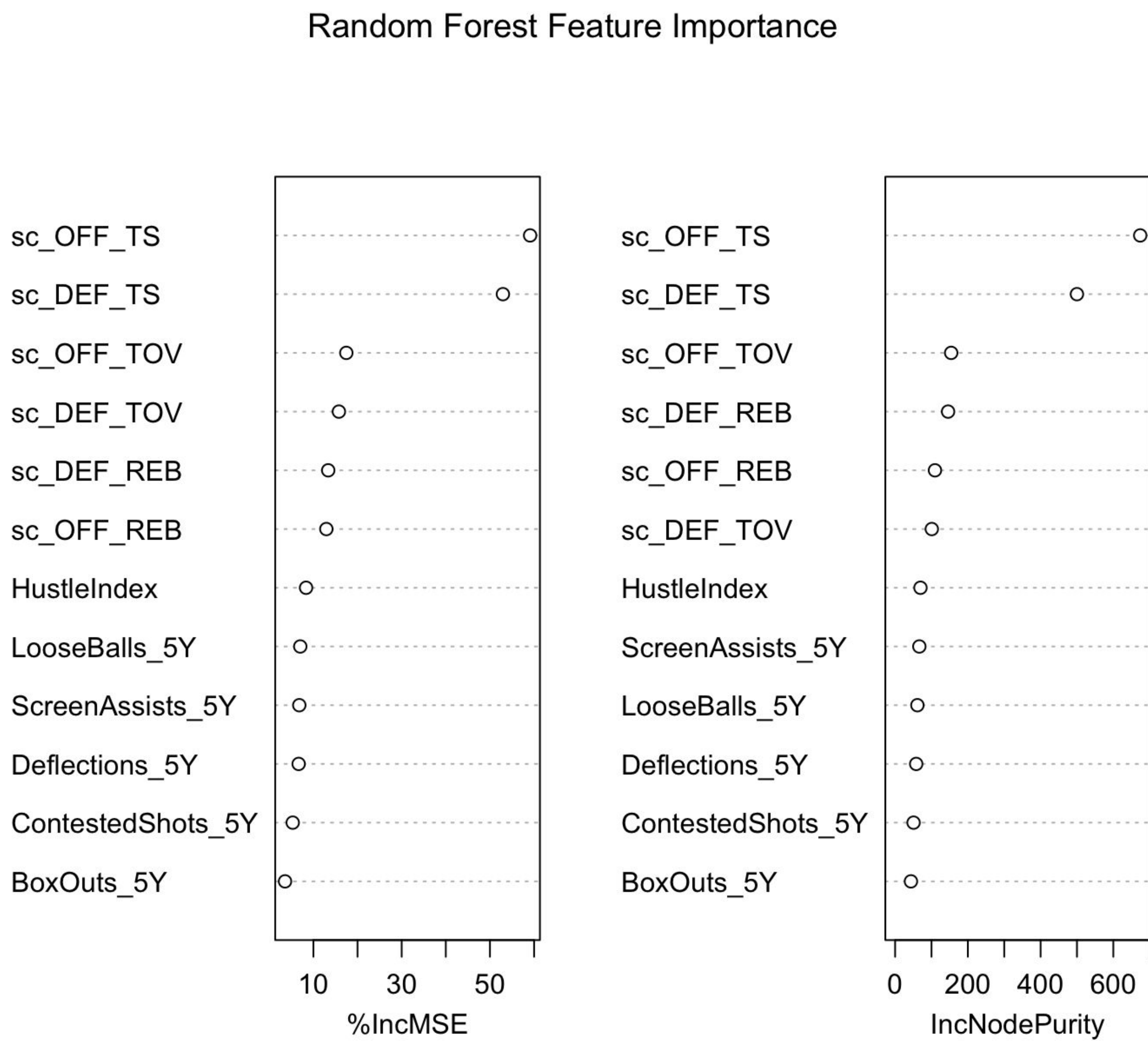
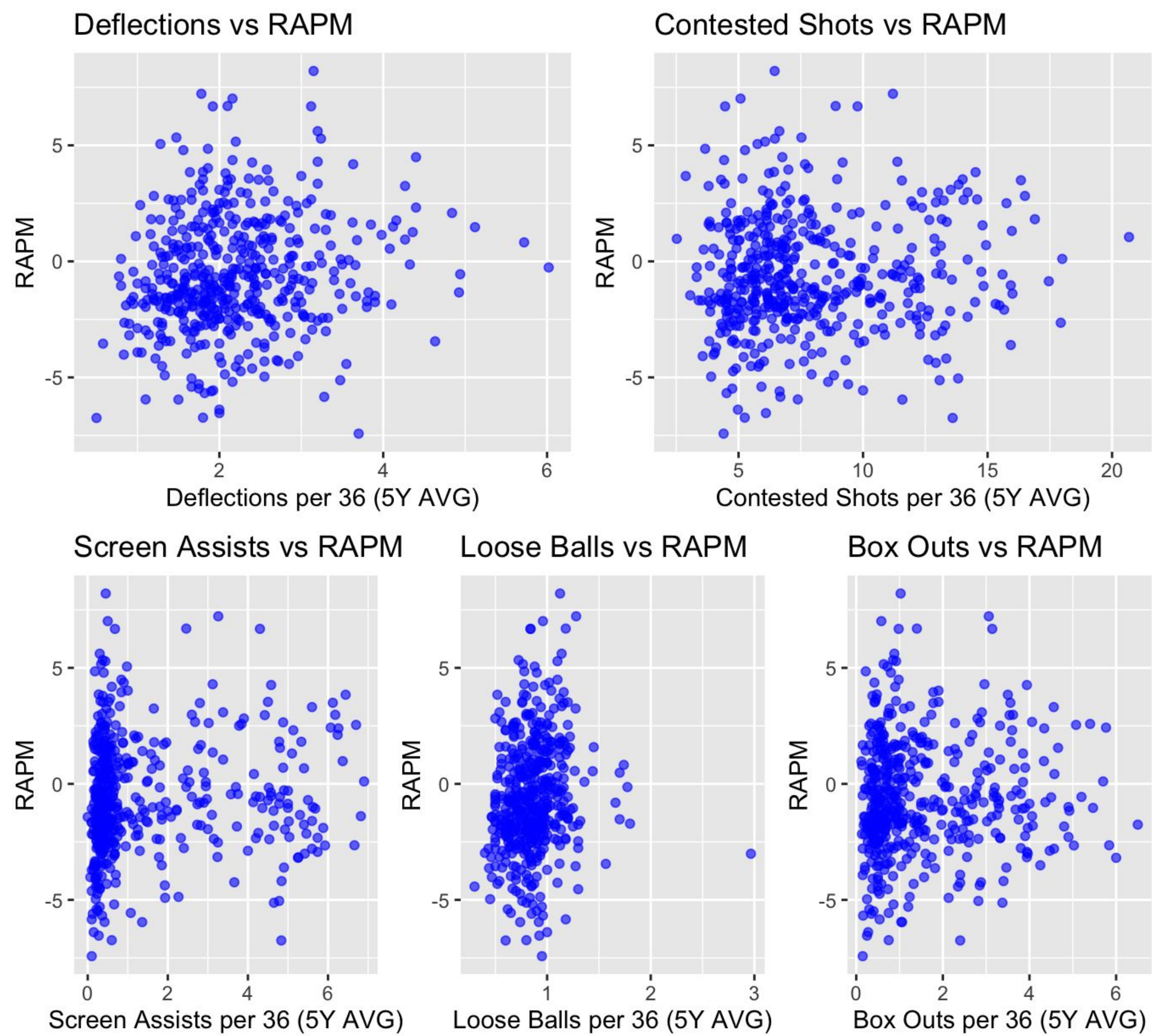
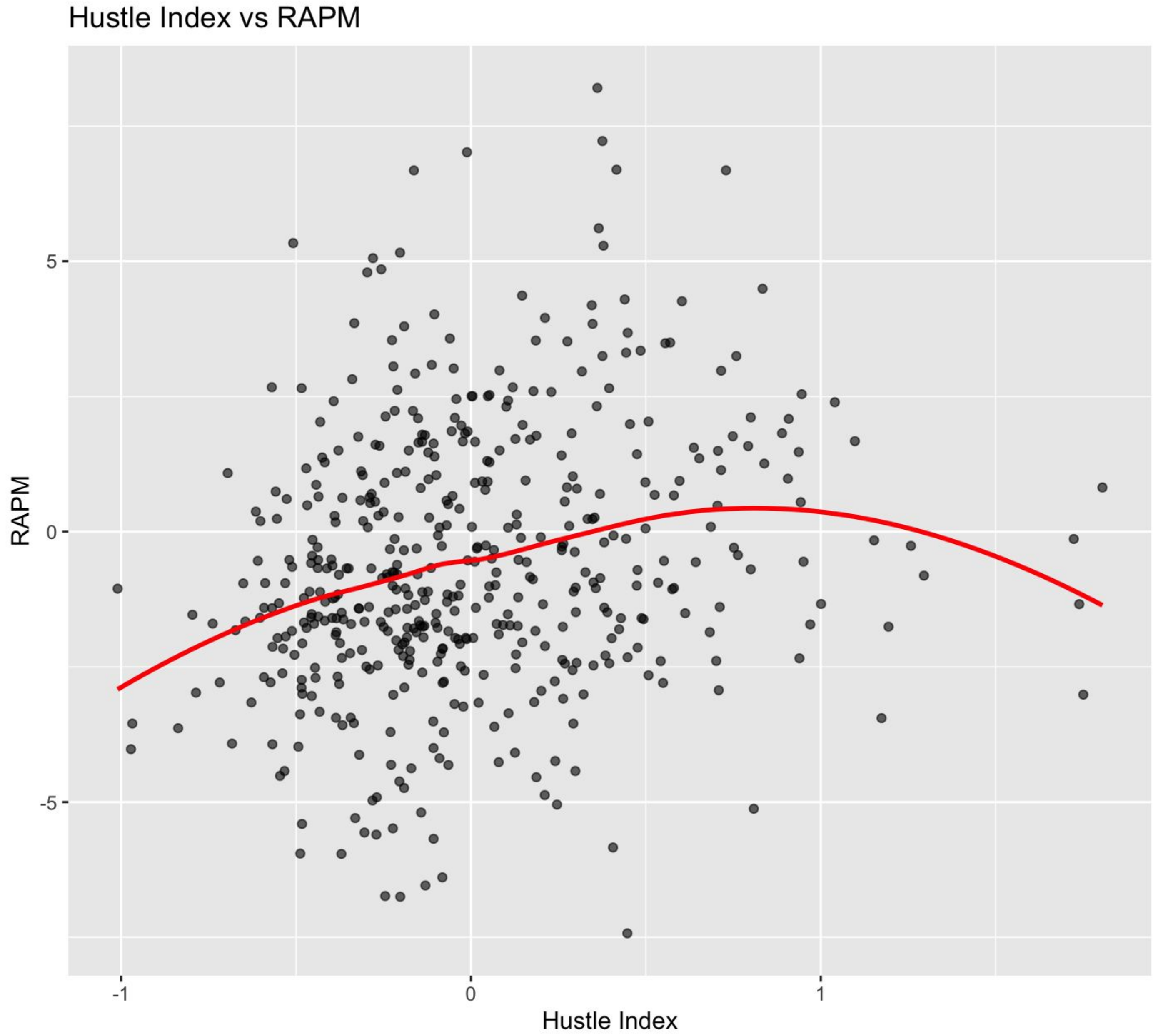
Machine Learning Model

- Use Decision Trees and Random Forest Models to evaluate the marginal importance of Hustle Index and the rest of the hustle stats when compared to the 6 Factors of RAPM.

Results

Top 5 5Y RAPM (2024)			
Rank	Player	RAPM	HI
1	K. Leonard	8.20	.36
2	J. Embiid	7.22	.38
3	J. Tatum	7.02	-.01
4	G. Antetokounmpo	6.69	.42
5	N. Jokić	6.68	.73

Top 5 Hustle Index (2019-2024)			
Rank	Player	RAPM	HI
1	Gary Payton II	.82	1.81
2	Sekou Doumbouya	-3.01	1.75
3	Nerlens Noel	-1.34	1.74
4	Paul Reed	-.14	1.72
5	Jarred Vanderbilt	-.81	1.3



- There's only a weak relationship between individual hustle stats or the combined Hustle Index and RAPM. High-impact players tend to hustle at similar rates to low-impact players.
- The overall relationship between Hustle Index and RAPM is weak and non-linear, showing diminishing returns on hustle, with a low 4% coefficient of correlation.
- While offensive and defensive efficiency are the primary predictors of RAPM, the Hustle Index still offers marginal predictive value according to Random Forest feature importance.
- Note: The analysis is slightly skewed because front-court players disproportionately perform plays like screen assists and box-outs.

Case Studies



RAPM: 8.20
Hustle Index: .36
Efficiency Over Hustle

Kawhi Leonard leads the league in RAPM, showing he is one of the most impactful players when on court. Leonard, however, has a Hustle Index that sits slightly above league average, suggesting that high-effort “hustle” plays are not the primary driver of his value. Instead, Leonard’s impact comes from elite defensive positioning, shot efficiency, and decision-making, showing that players can dominate RAPM without excelling in hustle metrics. This case illustrates how hustle contributes to impact in some players but is not a universal requirement for elite performance.

Alex Caruso has a reputation as a high hustle, impact role player. With a recorded RAPM of 4.49, and a strong Hustle Index of .83. Caruso’s value stems from the very actions hustle stats capture, creating turnovers and contesting shots. These two cases showcase the true gap in the league. Role players show their impact on the court through high effort plays, while the leagues superstars dominate traditional metrics through their efficiency and decision making.



RAPM: 4.49
Hustle Index: .83
Hustle-Driven Impact

Conclusions

- Hustle Index is a **marginal** feature in relation to Regularized Adjusted Plus Minus.
- The linear correlation of 0.21 indicates only a **weak direct relationship** between hustle metrics and overall player impact.
- Random Forest modeling further supports this conclusion, as including Hustle Index increased predictive accuracy only marginally (from 88% to 89%), a **statistically insignificant improvement**.
- Random Forest feature importance indicates that while offensive and defensive efficiency dominate prediction of RAPM, Hustle Index still provides **marginal predictive value**.
- The true value of Hustle Index lies in differentiating and evaluating role players, whose **contributions are often grounded in consistent high-effort plays**.
- Future research should further explore this relationship, focusing on the **role of hustle in sustaining on-court production** among non-star players.

Sources

https://www.nbarapm.com/datasets/six_factor
<https://www.nba.com/stats/players/hustle>

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